

Heaven's Light is Our Guide
Rajshahi University of Engineering & Technology
Department of Computer Science & Engineering

Lab Manual

Course Code: **CSE 4204**
Course Title: Sessional based on CSE 4203

Module 1: [Regression]

(for Week 1, Based on CO2)

Platform: Python with Anaconda Jupyter Notebook/other platform

Problem #1.[Play with Simple Dataset] Practice with simple dataset: **[BL:P2]**

- i. Create a sample dataset with one input feature and one output using Excel. Save the file in .csv format. Example House-Price, Hours of study-CGPA etc.
- ii. Open the file in python using Jupyter Notebook.
- iii. Display the rows and columns of the dataset
- iv. Train a model to Implement Linear Regression using the dataset
- v. Predict the output of the model using with real data

Problem #2. Suppose a model can predict the sugar level of patients. The following table shows real sugar values and predicted sugar values of 7 patients. [P2]

Real	2.4	4.6	7.3	4.1	3.3	6.8	5.7
Predicted	2.2	3.9	6.7	5.2	3.2	6.1	6.2

For the model, write python code to find

- i) MSE
- ii) MAE
- iii) RMSE
- iv) F2-Score

Problem #3. Train a model for Simple Regression using the following **[BL:P3]**

- i) Use dataset (placement.csv)
- ii) Divide the dataset into train and test [test size=20% or 30%]
- iii) Use Linear Regression
- iv) Calculate evaluation matrix (MSE, MAE and F2-Score)
- v) Test the model with real data
- vi) Print Coefficient and intercept values
- vii) Plot input features and regression line
- viii) Discuss the performance of the model with reasons.

Problem #4. Train a model for Multiple Linear Regression using the following **[BL:P3]**

- i) Use dataset ('homeprices.csv')
- ii) Divide the dataset into train and test [test size=20% or 30%]
- iii) Use Linear Regression for Multiple Features

- iv) Calculate evaluation matrix (MSE, MAE and F2-Score)
- v) Test the model with real data
- vi) Print Coefficient and intercept values
- vii) Plot input features and regression line
- viii) Discuss the performance of the model with reasons.

Problem #5. Train a model for Polynomial Linear Regression using the following **[BL:P3]**

- i) Use your own dataset
- ii) Divide the dataset into train and test [test size=20% or 30%]
- iii) Use Linear Regression for Polynomial Features
- iv) Calculate evaluation matrix (MSE, MAE and F2-Score)
- v) Test the model with real data
- vi) Print Coefficient and intercept values
- vii) Plot input features and regression line
- viii) Observe the model performance by changing the degree.
- ix) Discuss the performance of the model with reasons.